Replication Guide

Bug Report Classification Tool — How to Reproduce Results

Step 1: Clone the Repository

```
git clone https://github.com/USERNAME/ISWE.git
cd ISWE
```

Note: Replace the URL above with the actual repository URL.

Step 2: Set Up the Environment

```
python -m venv venv
source venv/bin/activate    # macOS/Linux
pip install -r requirements.txt
```

Step 3: Verify the Datasets

Confirm the following CSV files exist in `lab1/datasets/`:

- `tensorflow.csv`
- `pytorch.csv`
- `keras.csv`
- `incubator-mxnet.csv`
- `caffe.csv`

These datasets are included in the repository. They originate from the ISE course GitHub repository (https://github.com/ideas-labo/ISE).

Step 4: Run the Experiment

```
cd lab1
python main.py
```

Expected runtime: 2-5 minutes. The script prints progress for each project as it runs.

Step 5: Verify the Results

After the script completes, check that the following files have been created:

- `results/raw_results.csv` — 750 rows of individual run metrics
- `results/summary_results.csv` — 25 rows of aggregated statistics
- `results/wilcoxon_tests.csv` — statistical test outcomes
- `results/figures/` — 5 PNG box plot images (one per project)

Expected Results

Due to the use of `random_state=run` in the train/test splits, results should be **exactly reproducible**. The summary table should show Logistic Regression as the best-performing classifier on 4 of 5 projects by F1 score, with SVM winning on TensorFlow. All classifiers except Random Forest should show statistically significant improvement over the Naive Bayes baseline ($p < 0.05$) on all projects.

Approximate Expected F1 Scores

Project	NB (Baseline)	SVM	Log. Regression	XGBoost
TensorFlow	0.000	0.707	0.685	0.672
PyTorch	0.000	0.508	0.537	0.439
Keras	0.003	0.642	0.650	0.608
MXNet	0.000	0.523	0.603	0.461
Caffe	0.000	0.273	0.360	0.339

Random Forest results are omitted from this table as it largely failed on this task (near-zero F1 on most projects). Full results including Random Forest are in the raw output files.

Troubleshooting

- **NLTK download error:** Ensure internet access on first run. NLTK needs to download the stopwords corpus.
- **ModuleNotFoundError:** Ensure you activated the virtual environment and ran `pip install -r requirements.txt`.
- **FileNotFoundError on datasets:** Ensure you are running the script from inside the `lab1/` directory.